

Anytime-Valid Conjoint Analysis: A Digital-Camera Case Study

Supplementary case study for *Anytime-Valid Inference in Conjoint Experiments* (Molitor, Gosciak, and Lindon, 2026)

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1 Introduction

The parent paper establishes anytime-valid e-values and confidence sequences (CSs) for the linear-probability (“AMCE”) model of conjoint analysis, and demonstrates their properties through simulation. This note complements those simulations with a case study on a real conjoint dataset: a discrete-choice survey of digital camera preferences. The goal is to show what the method looks like end-to-end on real data – in particular, how the confidence sequence and e-value for each attribute effect evolve as respondents accumulate, and how the final anytime-valid interval compares to the conventional fixed- n interval computed from the same fitted model.

All code, data, and figures for this case study live in `cameras_case_study/` and can be regenerated with `run_all.sh`; see the accompanying `README.md` for exact reproduction steps.

2 Data

We use the `camera` dataset distributed with the R package `bayesm` (Rossi, 2023), a panel conjoint survey of digital camera preferences from Allenby, Brazell, Howell, and Rossi (2014a, 2014b); see also Case Study 4 in Rossi, Allenby, and McCulloch, *Bayesian Statistics and Marketing*. The dataset contains 332 respondents, each of whom completed 16 choice tasks. In every task the respondent chose among 4 camera profiles plus an explicit outside (“none of these”) option, for 26,560 respondent-task-alternative rows in total. Each profile is described by four mutually-exclusive brand indicators (Canon, Sony, Nikon, Panasonic – the outside option has all four set to zero), five binary feature indicators (higher pixel count, higher zoom, video capability, swivel display, WiFi), and price in hundreds of US dollars.

We treat each of the 332 respondents as a cluster, exactly as in the simulations in the main paper, and fit

$$\text{chosen} \sim \text{canon} + \text{sony} + \text{nikon} + \text{panasonic} + \text{pixels} + \text{zoom} + \text{video} + \text{swivel} + \text{wifi} + \text{price}$$

as a linear probability model via `feols()`, clustering standard errors by respondent. Each non-intercept coefficient is an AMCE: the average marginal effect of that attribute on the probability a profile is chosen, relative to the outside option.

3 Method

Point estimates and cluster-robust (“CRV1”) standard errors come directly from `feols()`. Anytime-valid p-values and confidence-sequence radii are then computed from those same estimates using the e-value construction of the parent paper, with the reference degrees of freedom and the process index both set to $G - 1$ and G respectively, where G is the number of clusters observed so far – not the row count – since clusters, not rows, are the unit that accumulates sequentially here.

The tuning parameter g , which controls the shape of the confidence sequence, is fixed once in advance at the value that minimizes the CS radius at the dataset’s actual final size, $G = 332$ (close to the ~ 300 clusters expected going in). This gives $g^* = 39.87$. Critically, g^* is computed once from the planned final sample size and held fixed throughout sequential monitoring below; re-optimizing g as data accumulate would invalidate the anytime-valid guarantee, since g would then depend on how the monitoring unfolds.

All significance levels below use $\alpha = 0.05$.

4 Results

4.1 Confidence sequences over time

Figure 1 shows every attribute’s confidence sequence narrowing steadily as more respondents are observed, while remaining valid to inspect at any point along the way – there is no multiple-looks penalty for having watched the price effect firm up over the first hundred or so respondents before checking in on swivel display or WiFi.

4.2 e-values over time

Figure 2 shows the corresponding e-value trajectories. Every attribute’s e-value exceeds the $1/\alpha$ rejection threshold by the end of the sample, i.e. every AMCE in this study is detectably different from zero given the full sample. The number of respondents needed to first cross that threshold varies substantially across attributes, from 25 for price to 110 for the weakest brand effect (Panasonic) – under a third of the full sample even in the slowest case. The trajectories are not always monotonic once above the threshold (Panasonic and Sony both dip back below it after their first crossing); this is expected of e-processes and is harmless under the relevant decision rule, which is to stop and reject the very first time the threshold is crossed, rather than requiring it to stay crossed forever.

4.3 Fixed- n interval vs. confidence sequence, at the final sample size

Figure 3 makes explicit the price of anytime validity at a single, fixed sample size: the two methods share the same point estimate (they are computed from the same model fit), but the confidence sequence is visibly wider than the conventional interval for every AMCE. That extra width is exactly what buys the ability to monitor the study continuously, as in Figures 1 and 2, and still report a valid interval or rejection decision at whatever point the analyst chooses to stop – something the conventional interval does not license.

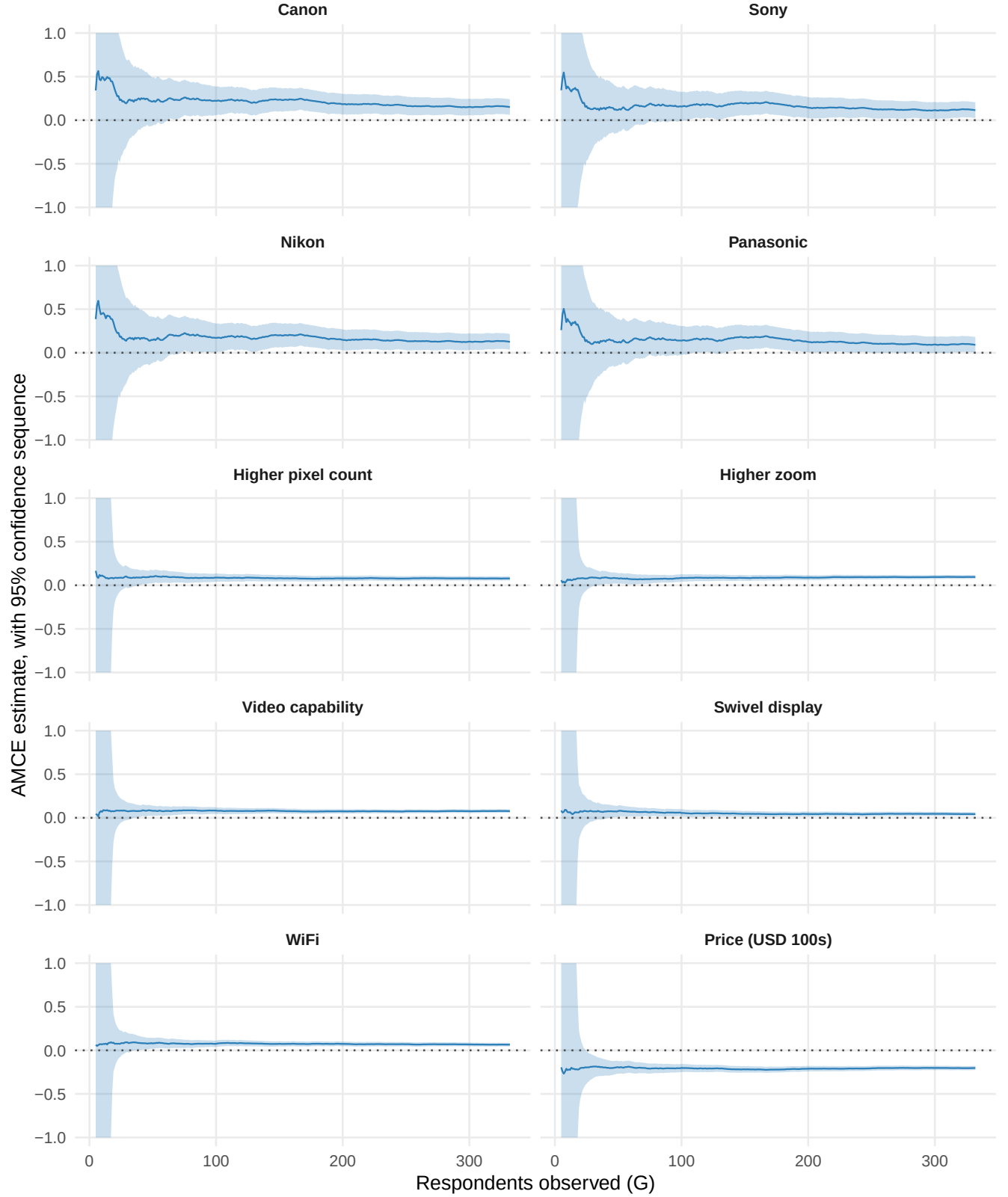


Figure 1: Anytime-valid 95% confidence sequence for each AMCE as respondents accumulate. Shaded regions clipped at ± 1 ; very early on (roughly $G < 18$) the true radius is unbounded, which is expected: a confidence sequence must sometimes be uninformative to guarantee valid coverage at every sample size, not just at a single pre-planned n .



Figure 2: Anytime-valid e-value for each AMCE as respondents accumulate (log scale). The dashed red line is $1/\alpha$, the threshold an e-value must cross to reject the null of no effect at level α ; the dotted grey line is $e = 1$ (neutral evidence).

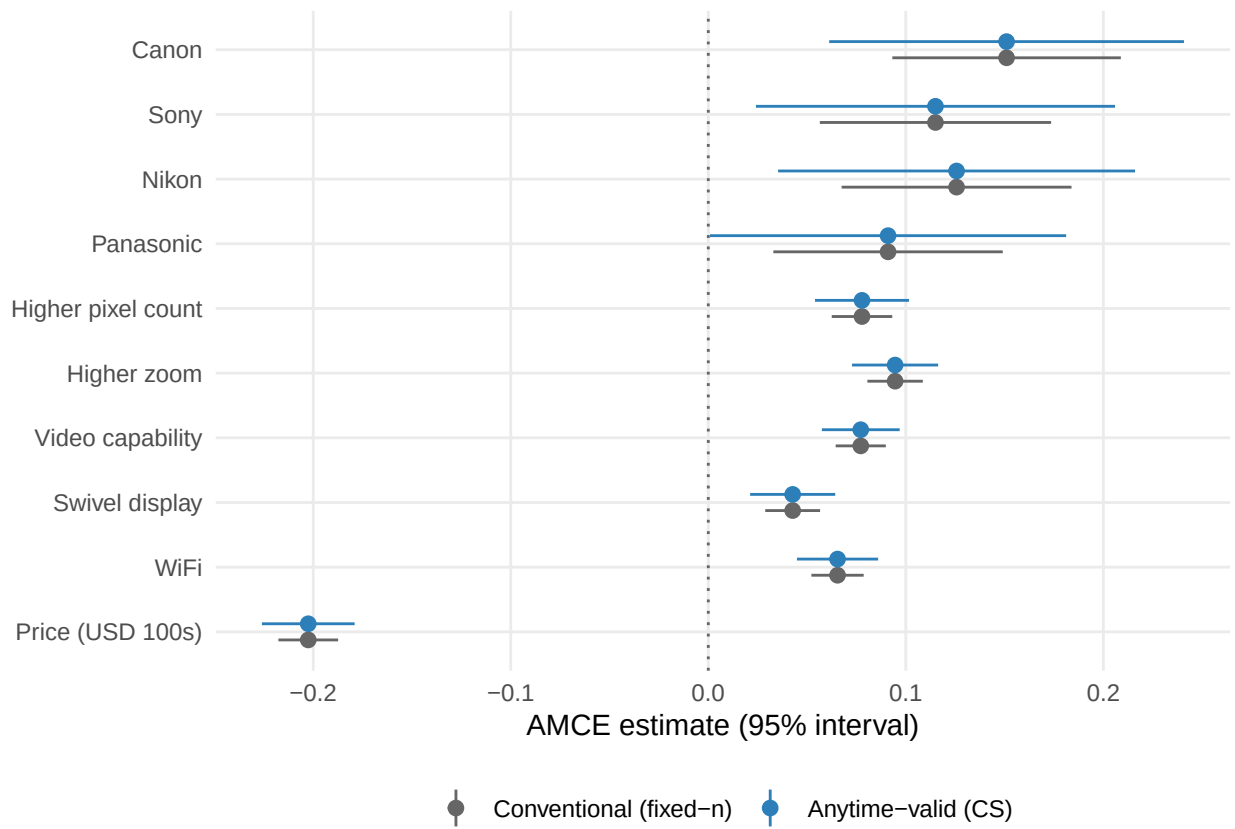


Figure 3: Final AMCE point estimate with a conventional fixed- n 95% confidence interval versus the anytime-valid 95% confidence sequence, both computed from the same final model fit ($G = 332$).

5 Discussion

On this real conjoint dataset, every attribute – four camera brands, five features, and price – has a confidence sequence that excludes zero and an e-value that comfortably clears the rejection threshold by the end of data collection, and the point estimates agree with intuition (all brands and features increase choice probability relative to the outside option; price decreases it). The case study illustrates the two costs and benefits of the anytime-valid approach side by side: wider intervals at any fixed sample size, in exchange for validity at every sample size, including whichever one an analyst actually decides to stop at.

6 References

- Allenby, G., Brazell, J., Howell, J., and Rossi, P. (2014a). Economic Valuation of Product Features. *Quantitative Marketing and Economics*, 12, 421-456.
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- Rossi, P., Allenby, G., and McCulloch, R. *Bayesian Statistics and Marketing*. Case Study 4.
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- Lindon, M., Ham, D., Tingley, M., and Bojinov, I. (2026). Anytime-valid linear models and regression adjusted causal inference in randomized experiments. *Journal of the American Statistical Association*. arXiv:2210.08589.
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